

Running Head: POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

Title: Muscle coactivation levels reflect a trade-off between metabolic cost and performance when responding to physical disturbances during upper limb posture control.

Abbreviated Title: Postural Coactivation Shows Cost-Performance Trade-off

Authors: Bhuvan Dubey¹, Philipp Maurus¹, Juliana Rancier¹, Jeromy Wong^{1,2}, Tyler Cluff^{1,3}

¹Faculty of Kinesiology, University of Calgary, Calgary, AB, Canada, T2N 1N4

²Department of Biomedical Engineering, University of Calgary, Calgary, AB, Canada, T2N 1N4

³Hotchkiss Brain Institute, University of Calgary, Calgary, AB, Canada, T2N 1N4

*Correspondence: bhuvan.dubey@ucalgary.ca or tyler.cluff@ucalgary.ca

Number of pages: 30

Number of figures: 9

Number of tables: 1

Number of words for abstract: 245

Number of words for introduction: 650

Number of words for discussion: 1500

Conflict of Interest: The authors declare no competing financial interests.

Acknowledgments: Hotchkiss Brain Institute, University of Calgary Faculty of Kinesiology

Abstract

Muscle coactivation refers to the simultaneous activity of muscles with opposing functions. Recent observations suggest that coactivation improves the nervous system's ability to respond to sensory feedback. However, muscle activity requires energy, and the relationship between this energy use and performance is not fully understood. This study investigated the trade-off between metabolic cost and performance under different levels of muscle coactivation. Healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances. The disturbances rapidly extended or flexed the elbow and displaced the participant's arm from the target, requiring them to return to the target. Participants first performed the task under self-selected levels of muscle coactivation, followed by biofeedback to adjust coactivation to [0.5, 2, 4, 8, 16]x their self-selected (1x) level. Metabolic cost was measured with gas respirometry, muscle activity using surface EMG, and behavioral responses using success rates. Trials were considered successful if participants return their arm back into the target within 400 milliseconds (ms) post-perturbation. Pilot data suggest that at self-selected levels of muscle coactivation, participant achieved a success rate of 30-50%. With higher coactivation levels, muscle responsiveness to disturbances increased, arm displacement was limited, and success rates was nearly 100%. However, increased coactivation levels through biofeedback resulted in an increased energy expenditure or metabolic costs. These finding suggests that participants may prioritize metabolic cost over performance at self-selected coactivation levels. This research provides insights into the role of muscle coactivation in sensory feedback response.

Introduction

During biceps curls, the biceps brachii contracts to lift the weight during the concentric phase, acting as the agonist muscle responsible for raising the weight. Meanwhile the triceps act as the antagonist muscle, and coactivate to stabilize the elbow, prevent excessive motion and maintain controlled movement during this concentric contraction. This concept is widely recognized for its use in controlled stability tasks including balance, jogging and changing posture (Vieira et al., 2019). However, new research indicates that muscle coactivation may not only stiffen muscles but also improve the nervous system's responsiveness to sensory input (Balshaw et al., 2018).

Beyond stiffness, muscle coactivation is closely linked to how the nervous system processes sensory feedback through spinal and brain circuits (Maurus et al., 2024). The short-latency response (SLR), which is quick spinal reflex, responds rapidly to muscle stretch with a time delay of 25–30 milliseconds (Maurus et al., 2024). Then the long-latency response (LLR) starts at 50–100 milliseconds to provide adaptable motor control through cortical processing (Maurus et al., 2024). While SLR reflects passive muscle properties, but LLR includes descending control, which may be influenced by coactivation (Maurus et al., 2024). However, the way coactivation modulates feedback responses in dynamic tasks remains unclear and requires further investigation.

Most of the current research on coactivation is based on isolated muscle experiments or preclinical models. In these investigations, scientists use calcium solutions or nerve stimulation to control muscle activity (Gebel et al., 2019). The mechanical properties of stiffness and resistance to stretching can be measured by the researchers through changes in calcium concentration (Gebel et al., 2019). The studies offer valuable knowledge, but they do not track neural feedback that

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

people experience when performing real-world activities which need motor control and sensory processing (Kesar et al., 2019). These experiments lack complete understanding of coactivation effects on movement because they do not track neural feedback.

Research involving human participants mainly studies static conditions because coactivation and stiffness show direct relationships in these settings. Under static muscle activation muscle stiffness develops because of actin-myosin cross-bridges yet this relationship does not apply when studying dynamic movements (Vieira et al., 2019). Movement creates unpredictability which makes actin-myosin binding occur probabilistically thus stiffness cannot be measured only by muscle activation levels (Vieira et al., 2019). Researchers continue to use static-condition results for dynamic tasks even though this practice creates important knowledge gaps about real-world movement coactivation mechanisms.

Another important knowledge gap in the literature is the lack of focus on sensory feedback that needs further investigation. The majority of coactivation studies using surface electromyography (EMG) do not evaluate how this coactivation affects disturbance responses and sensory input processing. Coactivation reduces movement variability and boosts precision (Balshaw et al., 2018), but its effect on the nervous system's ability to process sensory data and handle perturbations remains unclear. Sensory feedback plays an important role in guiding movement control, making this a critical area to investigate. The trade-off between metabolic cost and performance is also poorly understood. While enhanced coactivation improves motor control and responsiveness according to Peterson & Martin (2010), it increases energy requirements which may reduce endurance for long activities. This study examines how muscle coactivation affect the trade-off between performance and metabolic cost during upper limb posture control with mechanical disturbances, an area requiring further research.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

The hypothesis is that there is a trade-off between muscle coactivation, metabolic cost, and task performance. Higher coactivation levels may improve responsiveness to perturbations and improve performance (lower error rates) but it might come at the expense of increased metabolic cost. On the contrary, lower coactivation may save energy expenditure but reduce performance. Coactivation enhances stability and control, yet its energy impact in rapid, precise movements is unclear. By exploring this balance, the research seeks to provide insight into how the nervous system adjusts coactivation for optimal stability and energy efficiency, with findings to inform future analyses and understand motor control.

Methodology**Participant Characteristics**

Twelve right-handed participants (7 male and 5 female) were recruited from the University of Calgary. The age, height, and weight of the participants were 21.92 ± 5.96 years, 173.17 ± 11.26 cm and 74.16 ± 17.80 kg, respectively (mean $\pm$ SD). Only participants who were free of self-reported neurological or musculoskeletal conditions were included in the study, and participants had normal or corrected vision. All participants signed consent forms before doing the experiment, and the study protocol was approved by the University of Calgary Conjoint Health Research Ethics Board (CHREB). The entire data collection session was about 1.5-2 hours long, including preparation, and it was completed as a cross-sectional (single timepoint) study. Each participant was paid \$20 CAD at the end to compensate for travel and time expenses.

Apparatus set-up

The Kinarm exoskeleton robot (Calgary, AB, Canada) was used as the primary apparatus for this experiment, designed to restrict arm movements to a horizontal plane. Participants' arms were placed into the troughs of the device, which permitted smooth lateral motion while preventing

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

vertical displacement. Adjustments were made to the chair height to align participants' arms with the exoskeleton, ensuring feet rested flat on the footrest and backs were supported comfortably. A shoulder indicator light ensured shoulders remained level, aligned with the acromion process, while armrests were tuned to hold arms comfortably with elbows slightly bent. The robot's joints were synced to match participants' shoulder, elbow, and wrist positions. The exoskeleton enabled nearly frictionless movement at these joints and could apply controlled forces to the shoulder or elbow while tracking motion. Targets (0.75 cm in size) and a white cursor (0.5 cm radius), tied to the index finger, were shown on a TV screen connected to the Kinarm. A shield was used to block the view of the hands, forcing reliance on the screen's cursor for feedback.

The Cosmed K5 wearable metabolic system was also utilized to measure metabolic activity. The K5 mask, covering the nose and mouth, was fitted snugly, and participants were asked to limit talking to avoid skewing breathing data. This system captured VO₂ and VCO₂, with specific timings recorded for analysis of metabolic cost using the Brockway equation (1987), which converts VO₂ into energy expenditure.

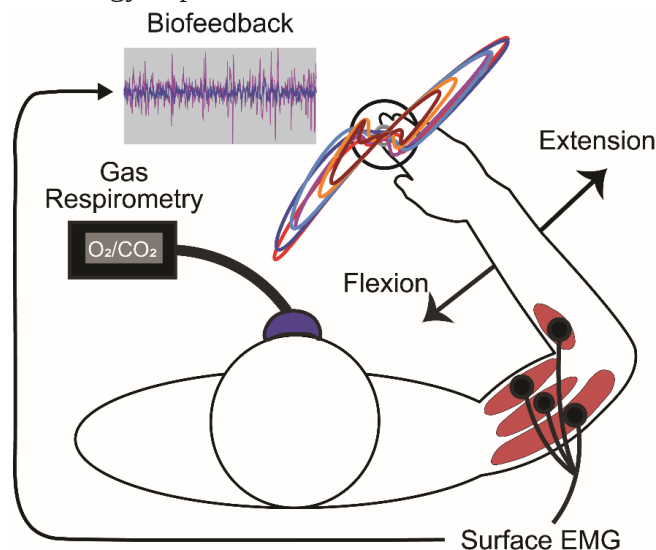


Figure 1. Schematic of the task protocol setup. This figure illustrates the experimental setup for the task protocol, depicting a side view of the participant's arm with the elbow positioned for extension (outward arrow) and flexion (inward arrow) movements induced by the Kinarm exoskeleton robot. Concentric circles at the right hand represent exemplar arm trajectories for each

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

coactivation level. A biofeedback signal, shown as a wavy line inset, indicates the real-time muscle activation feedback used to adjust coactivation to the specified levels. The VO₂ mask, colored blue, is positioned over the participant's face to measure metabolic cost via gas respirometry, while red areas on the arm highlight the four muscles (brachioradialis, biceps, triceps lateralis, and triceps longus) monitored using surface EMG.

Task Protocol

The experiment was structured into three distinct tasks, each with specific procedures outlined below.

Task 1: Self-Selected Coactivation Condition with Mechanical Disturbances

Before starting, baseline resting breathing rates were recorded after participants rested for five minutes, capturing VO₂ and VCO₂ with the Cosmed K5. This task, labeled condition 1x (self selected), began with a small circular target at the screen's center. Participants guided the white cursor into the target and maintained a steady arm posture. The Kinarm exoskeleton robot then introduced a mechanical disturbance, rapidly extending or flexing the elbow to displace the cursor from the target. Participants were required to return the cursor to the center within 400 milliseconds (ms). If the return occurred within the time limit, the target turned green. If the return was too slow, the target turned blue, signaling that the participant did not meet the time requirement for that trial. Short breaks were provided between trials to prevent fatigue. Time durations for the K5 metabolic mask were recorded at the beginning and end of condition 1x and noted in a separate Excel datasheet.

Task 2: Adjusted Coactivation Conditions with Biofeedback and Increasing Intensity

EMG data saved from task 1 were exported into MATLAB. Then baseline biceps EMG activity data were normalized and averaged using MATLAB and multiplied the recorded values by the coactivation levels (0.5x, 2x, 4x, 8x, and 16x) to determine reference EMG for each condition which was used for task 2. A circular target appeared on the screen, and the cursor was positioned at its center. A magenta slider displayed the participant's current muscle activation, while a green

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

biofeedback box indicated the required activation level for the trial. The slider's position reflected the biceps contraction level. Higher muscle activation raised the slider, and lower activation lowered it. When the slider aligned within the green rectangular box, indicating the correct level of biceps contraction for the condition (0.5x, 2x, 4x, 8x, or 16x relative to the self selected 1x level), the robot applied a torque force, displacing the arm from the target. Participants then had to return the cursor to the circular target as quickly as possible. If the cursor returned within 400 ms, the target turned green. If the return did not occur within 400 ms, the target turned blue, indicating a failed trial. The task increased in difficulty over time, with conditions progressing from 0.5x to 16x, and continued until participants could no longer complete the trials due to fatigue or inability to maintain the required coactivation level. Time durations for the K5 were logged at the beginning and end of each condition and recorded in the Excel datasheet.

Task 3: Normalization Task for Establishing Baseline Muscle Activity

The third task, referred to as the normalization task, involved holding the cursor in the center of a circular target for approximately two minutes while participants relaxed under a small load. This task was designed to establish a baseline for muscle activity under minimal effort, providing a reference point for normalizing EMG data across participants and conditions. No VO₂ time was measured during this task. Each condition in Tasks 1 and 2 consisted of 80 trials, totaling 480 trials per participant, though the number of completed trials varied due to differences in endurance. Task 3 included six trials per participant to ensure sufficient data for normalization purposes.

Kinematics

Movement data were collected from the Kinarm system to evaluate arm kinematics throughout the tasks. Elbow joint angles were captured at a sampling rate of 1 kHz and processed with a second-order, dual-pass Butterworth filter using a 30 Hz cutoff frequency to remove noise. The change in

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

elbow angle, referred to as delta (measured in degrees (deg)), was calculated at the onset of each trial to capture initial joint positioning. Movement time (recorded in milliseconds) was defined as the duration from the cursor exiting the target until its return, with success indicated by a return within 400 ms, earning a green target. Data processing in MATLAB R2019b yielded precise measurements of movement time, elbow angles, and peak elbow angle.

Muscle Activity

Surface EMG was employed to monitor muscle activity during the tasks. Prior to attaching electrodes, the skin was prepared by shaving with disposable razors if necessary and cleaned with alcohol swabs to enhance signal quality. Bipolar surface electrodes (DE 2.1 Single Differential Electrode) (Delsys Bagnoli, Delsys Inc, Natick, MA, USA) were positioned over the brachioradialis, triceps lateralis, biceps and triceps longus to record activity from these upper limb muscles. A reference electrode was placed on the lateral epicondyle, and the EMG system, connected to a computer, recorded data at a sampling rate of 1 kHz. Signals were amplified with a gain of 1000 and subjected to a band-pass filter (20–450 Hz, dual-pass, 3rd-order Butterworth) before full-wave rectification to isolate muscle activity. Normalization was achieved by having participants hold reference loads of ± 1 Nm to flex and extend the elbow, maintained for 15 seconds in two directions (Task 3). These normalized EMG values were expressed in normalized units (nu), enabling consistent comparisons across participants by accounting for individual muscle strength differences. For analysis, EMG data from all trials were aligned to the onset of the robot's disturbance. Muscle activity was averaged for each muscle within defined time windows: SLR (25–50 ms), and LLR (50–100 ms). Only elbow monoarticular and biarticular muscles were looked at because they directly influence elbow stability and movement. Shoulder muscles were left out to focus only on the primary joint of interest and avoid adding complexity from upper arm

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

dynamics. To isolate the perturbation-evoked response (ΔEMG , in normalized units), the average activity from unperturbed trials was subtracted from that of perturbed trials. The ΔEMG was calculated for each trial, averaged per participant, and then averaged across participants for each condition. Profiles with age, height, and weight were created for new participants, and all EMG data were processed and analyzed using custom MATLAB R2019b scripts.

Data Analysis and Outcome Measures

All analysis was handled in MATLAB R2019b. Mean metabolic rates and standard deviations were calculated, and metabolic cost was determined from VO_2 data using the Brockway equation (1987). Behavioral responses were modeled with an exponential fit, using the equation:

General model Exponential Fit (1 term):

$$f(x) = a * e^{b*x} \quad \text{(Equation 1)}$$

In this equation, x represents the independent variable (coactivation level), and $f(x)$ is the dependent variable (behavioral response). The parameter ‘ a ’ is the initial value (pre-exponential factor), while ‘ b ’ is the growth/decay constant. If $b > 0$, it indicates exponential growth, and if $b < 0$, it represents exponential decay. The base ‘ e ’ is the natural logarithm constant (~ 2.71828). Parameters were extracted in MATLAB to quantify behavior across conditions. Figures were also generated in MATLAB to visualize trends. The study gathered data on metabolic cost through gas respirometry, checked success rates and speed in dynamic tasks, and tracked muscle coactivation with EMG to see how these changed with different coactivation levels.

Results

Kinematic Responses Across Coactivation Levels

The elbow angle data were recorded during the extension and flexion tasks to capture movement across coactivation levels (0.5x, 1x, 2x, 4x, 8x, 16x). Two figures present these measurements, one showing the progression over time and the other summarizing the peak values observed.

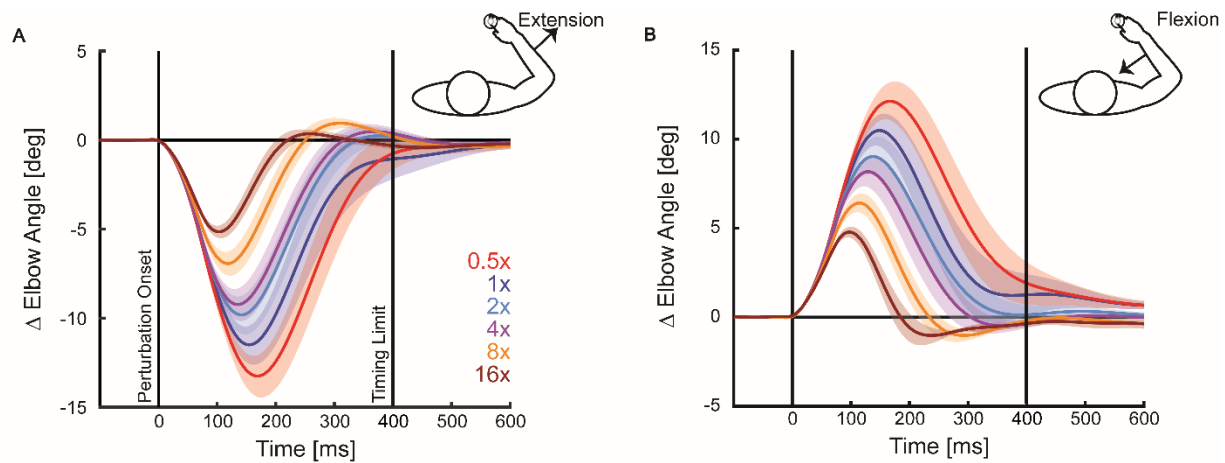


Figure 2. Average elbow angle of participants over time for the extension (A) and flexion tasks (B). This provides a detailed view of elbow angle changes across all conditions. Each coactivation level is marked by a specific line color: 0.5x in light red, 1x in dark blue, 2x in light blue, 4x in purple, 8x in dark orange, and 16x in dark red. The shaded areas show variability between participants.

The angle starts at 0 degrees when the cursor is on the target and changes after the robot applies a disturbance. For the extension task, the elbow angle reaches about -14 degrees at 0.5x (light red) and drops to around -5 degrees at 16x (dark red). For the flexion task, the angle peaks at about 13 degrees for 0.5x (light red) and falls to around 5 degrees for 16x (dark red). Higher coactivation levels (e.g., 8x, 16x) showing smaller changes in elbow angle. The shaded areas show how much variation there was between participants and are widest at smaller coactivation levels (e.g., 0.5x) and narrowest at 16x. The elbow moves the most when coactivation is low and the least when

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

coactivation is high. The flexion and extension patterns look similar, but the direction of movement is opposite.

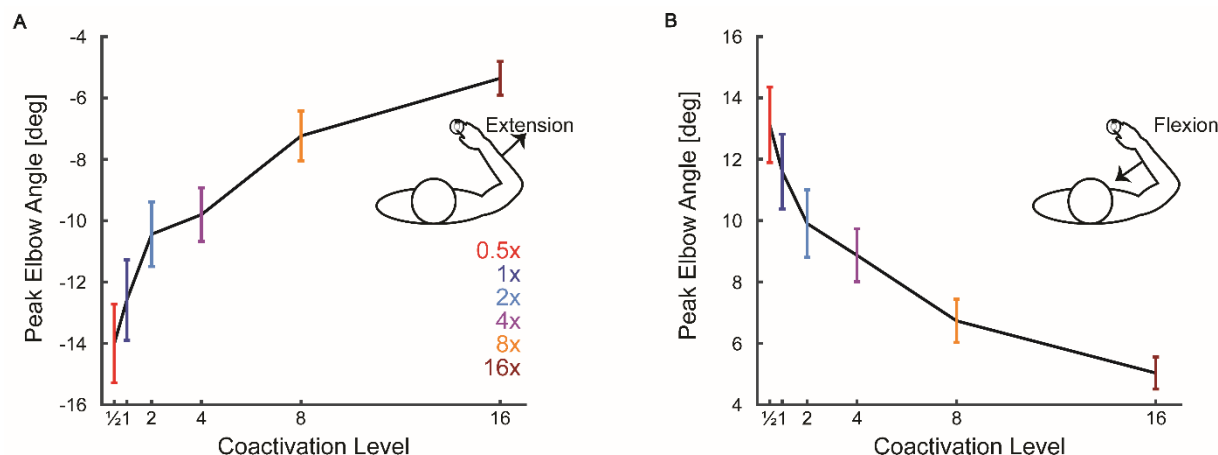


Figure 3. Summary of peak elbow angle for extension (A) and flexion (B) protocols. Peak elbow angles for each coactivation level are presented, maintaining the same color scheme as Figure 2. The error bars show variability across participants.

Figure 3 lists the peak elbow angle values recorded across all conditions. The extension and flexion tasks follow the same pattern, but the angles are in opposite directions. In the extension task, the peak elbow angle is -14.5 degrees at 0.5x and -5 degrees at 16x. In the flexion task, the peak elbow angle is 13 degrees at 0.5x and 5 degrees at 16x. There is more variation (larger error bars) at lower coactivation levels by approximately ± 1 degrees at 0.5x compared to ± 0.5 degrees at 16x. As muscle coactivation increases, the peak elbow angle becomes smaller, so the arm moves less from its starting position. The relationship between coactivation and peak elbow angle follows a clear exponential trend (Table 1), with movement decreasing as coactivation increases.

Additionally, movement time was measured to determine the duration required for participants to return the cursor to the target following the disturbance. Figure 4 shows that muscle coactivation facilitates faster movements.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

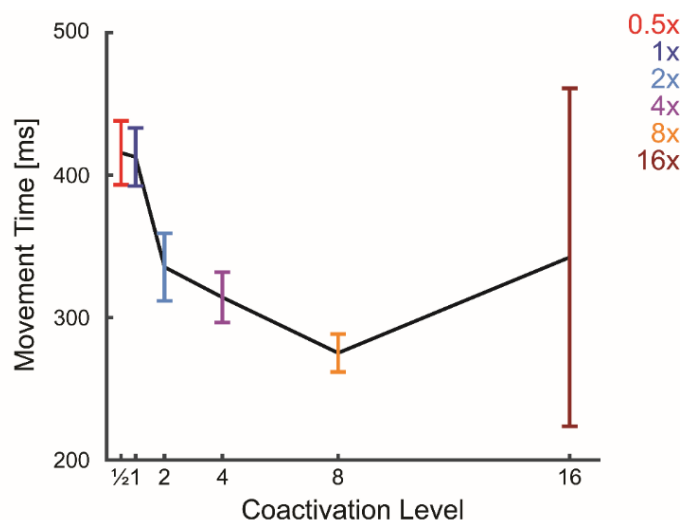


Figure 4. Average movement time for participants at different coactivation levels, with colors consistent with prior figures. Error bars indicate variability across participants.

Movement time decreases as coactivation increases, reaching the lowest point at the 8× coactivation level. The recorded times are approximately 420 milliseconds at 0.5x and 280 milliseconds at 8x. At the highest level of coactivation, movement time increases again to about 340 milliseconds. The largest reduction in movement time occurs between the lowest and intermediate coactivation levels. At the highest level of coactivation, it shows the most variation in movement time (approximately ± 250 milliseconds at 16x).

Muscle Activity Responses Across Coactivation Levels

Electromyography (EMG) data were gathered to assess muscle activity during the extension and flexion tasks across coactivation levels. Four figures present these measurements, two for each task, detailing the responses of four muscles.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

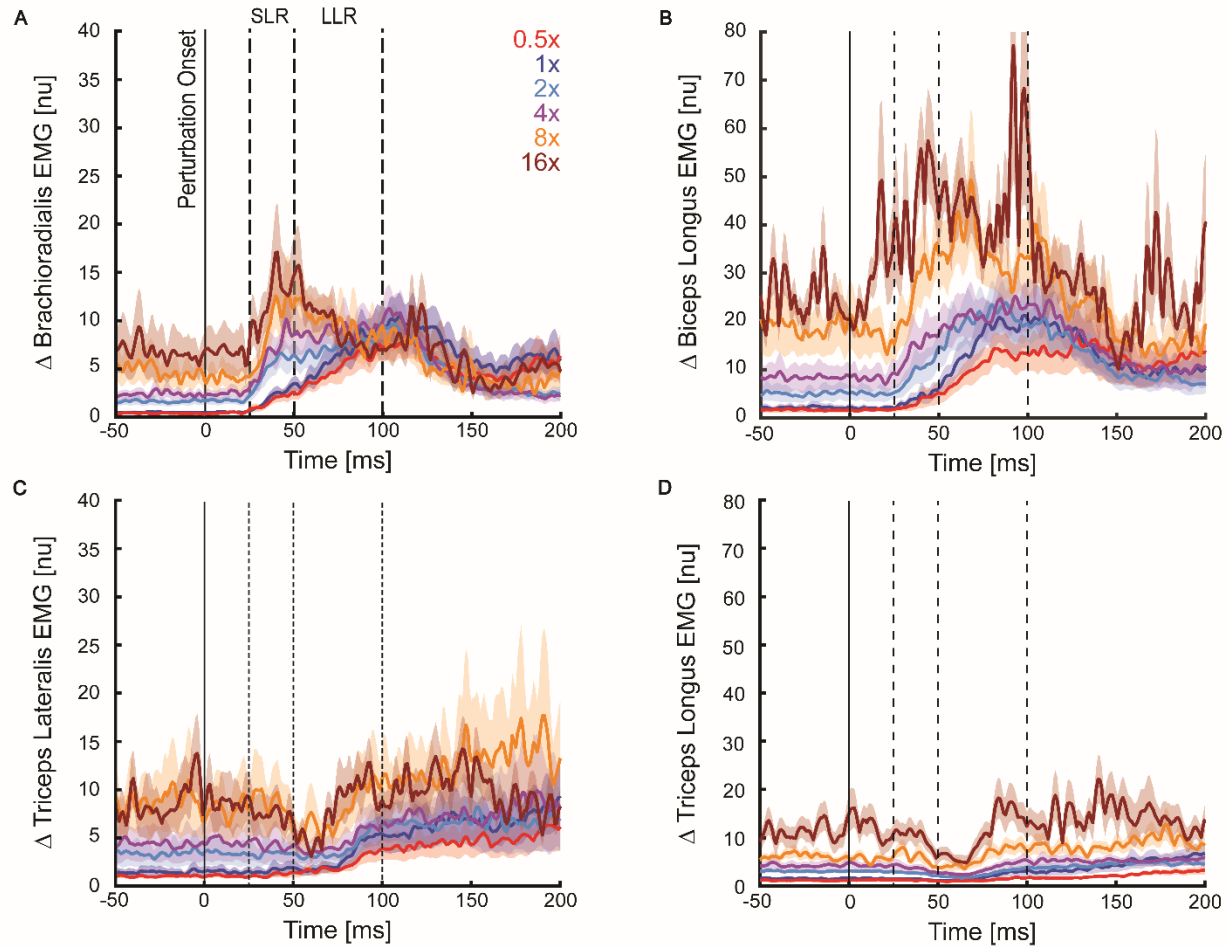


Figure 5. Normalized EMG activity for group average during the extension task is plotted for four muscles across coactivation levels, with colors consistent with previous figures. During the extension task, the brachioradialis (A) and biceps longus (B) act as the agonist muscles, while the triceps lateralis (C) and triceps longus (D) act as antagonist muscles. Perturbation onset is at 0 ms (the start of the trial). SLR is shown by the dotted line between 25-50 ms and LLR is shown by the dotted line between 50 – 100 ms. The shaded areas show variability between participants.

When coactivation levels increased, the agonist muscles had a stronger reaction, shown by higher EMG activity. The biceps longus showed the biggest increase in activation, while the brachioradialis also became more active, peaking early in the movement. On the other hand, the antagonist muscles were more inhibited as coactivation increased. Both the triceps lateralis and longus had lower EMG activity, meaning they were more effectively suppressed. Overall, higher muscle coactivation increased agonist activity while suppressing antagonist muscles.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

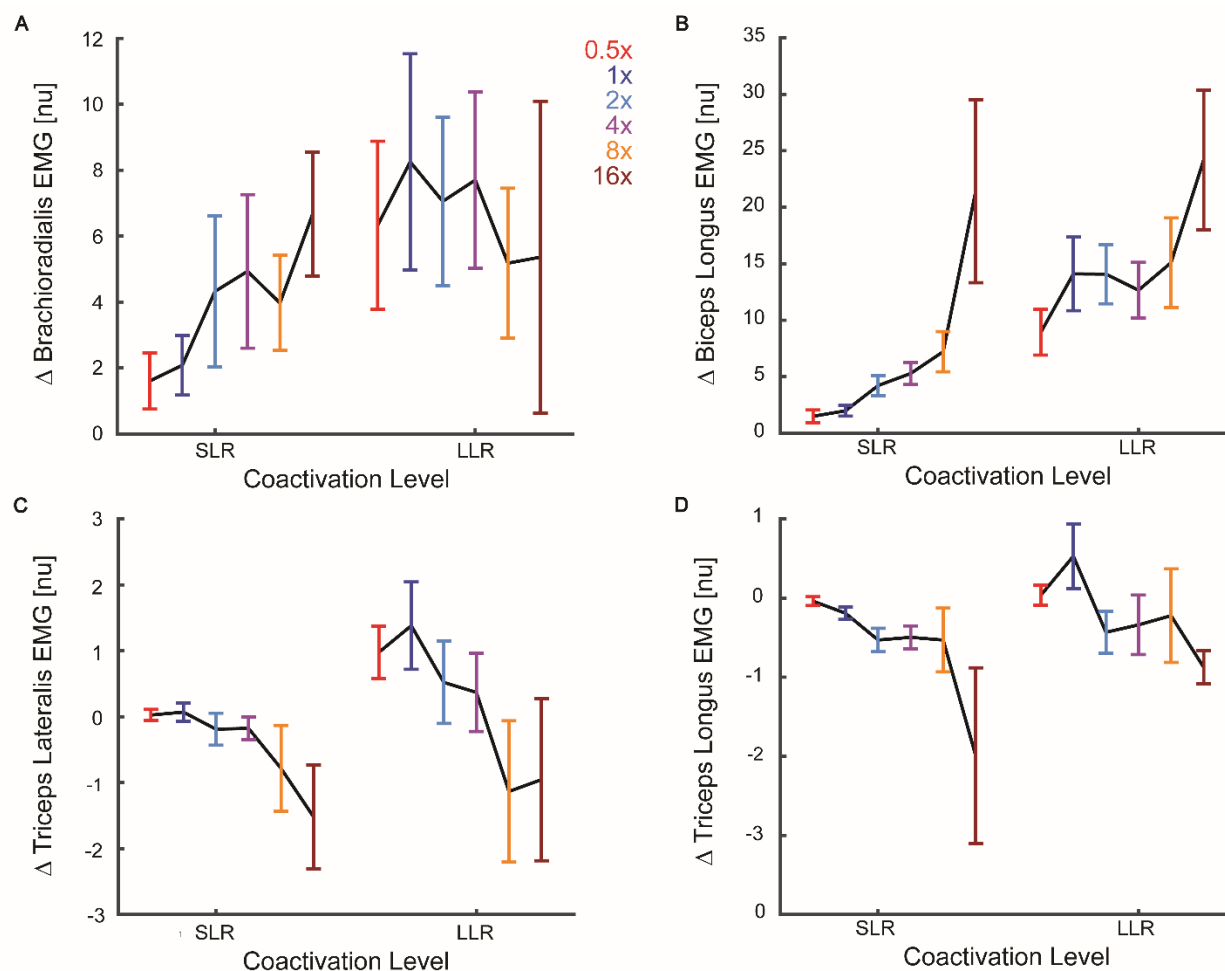


Figure 6. Summary of muscle activity for the SLR and LLR epochs across conditions for the extension task. SLR is shown on the left and LLR is shown on the right. Error bars indicate variability across participants.

EMG activity during the extension task is divided into SLR (25 to 50 ms) and LLR (50 to 100 ms) epochs, using the same color scheme. During the extension task, the agonist muscles, the brachioradialis (A) and biceps brachii (B), showed increased activation as the movement progressed from the short-latency response (SLR) to the long-latency response (LLR), with the strongest activation seen in the LLR phase. For brachioradialis, activity is approximately 1.5 nu at 0.5x and 6.5 nu at 16x in SLR, and 6.2 nu at 0.5x and 5.5 nu at 16x in LLR. Similarly for biceps

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

294 longus, the activity is about 2 nu at 0.5x and 20 nu at 16x in SLR, and 8 nu at 0.5x and 22.5 nu at
295 16x in LLR. In contrast, the antagonist muscles, the triceps lateralis (**C**) and triceps longus (**D**),
296 showed a decrease in activation during the SLR phase (went from 0 nu to -1.5 nu for triceps
297 lateralis and 0 nu to -2 nu for triceps longus from 0.5x to 16x coactivation), meaning they were
298 inhibited early in the movement. Their activity slightly increased again in the LLR phase for 0.5x
299 and self-selected levels but more coactivation resulted in the LLR phase stretch activity to drop (-
300 1 nu for triceps lateralis and -0.8 nu for triceps longus at 16 x). The error bars show some variability
301 between participants, but the overall trend remained consistent where the agonist muscles became
302 more active over time, while antagonist muscles were suppressed, especially in the early phase
303 (SLR) of the response.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

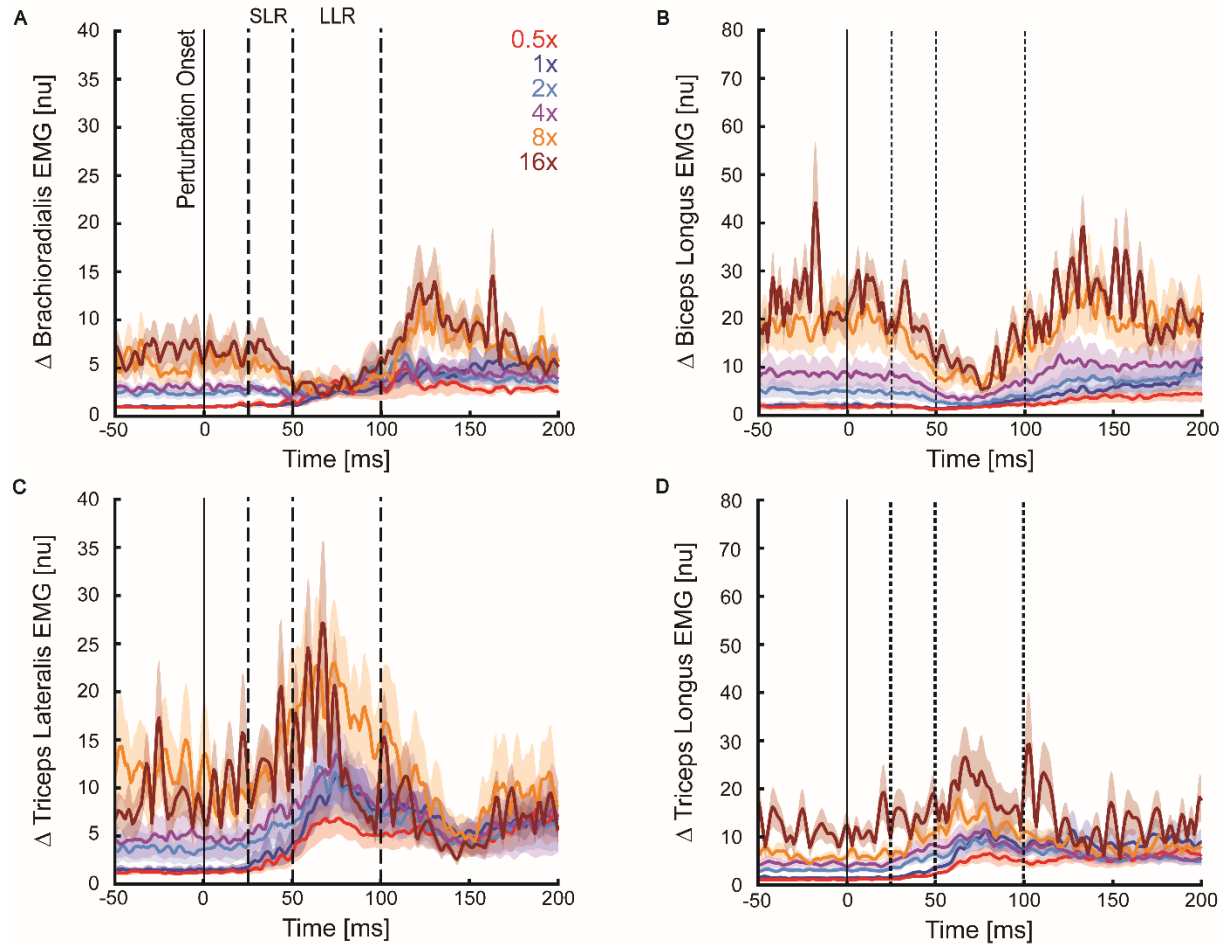


Figure 7. Normalized EMG activity for group average during the flexion task is plotted for four muscles across coactivation levels, with colors consistent with previous figures. During the flexion task, the brachioradialis (A) and biceps longus (B) act as the antagonist muscles, while the triceps lateralis (C) and triceps longus (D) act as agonist muscles. Perturbation onset is at 0 ms (the start of the trial). SLR is shown by the dotted line between 25-50 ms and LLR is shown by the dotted line between 50 – 100 ms. The shaded areas show variability between participants.

Muscle coactivation influences how strongly muscles respond to flexion perturbations. The agonist muscles (triceps lateralis and triceps longus) show greater activation as coactivation increases, with the triceps longus displaying the strongest response. These muscles react more strongly after the perturbation onset, particularly during SLR and LLR epochs. For triceps lateralis and triceps longus, stretch activity is lowest at 0.5x (light red line) and highest at 16x (dark red line). Triceps lateralis showed a stronger stretch activity compared to triceps longus. In contrast, the antagonist muscles (brachioradialis and biceps longus) show more inhibition as coactivation increases,

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

meaning they are less active when opposing movement. The shaded regions represent variability between participants, but the overall trend remains consistent, showing that higher coactivation results in greater agonist activity and more suppression of the antagonist muscles.

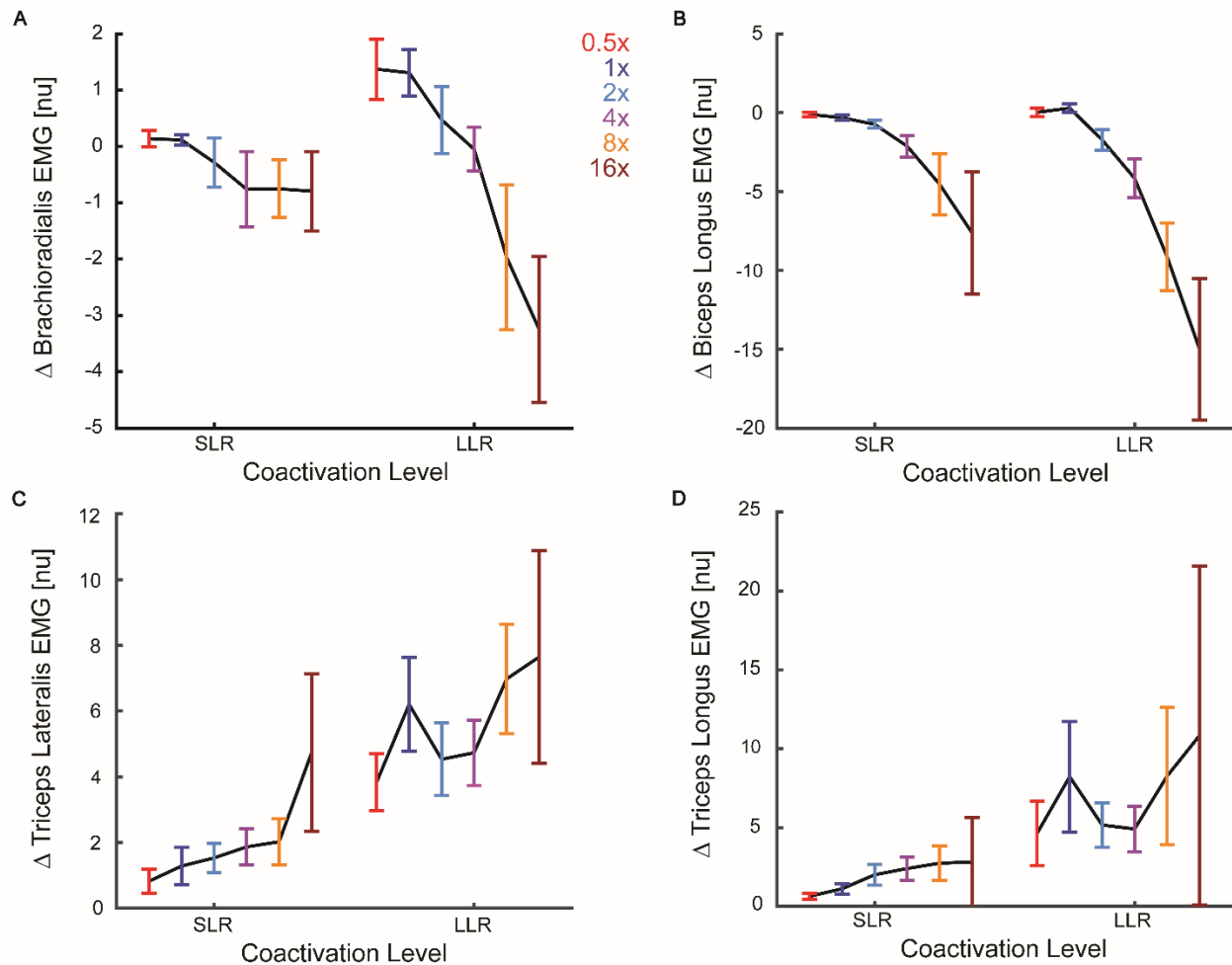


Figure 8. Summary of muscle EMG activity for the SLR and LLR epochs across conditions for the flexion task, maintaining the same color scheme as previous figures. During the flexion task, the brachioradialis (A) and biceps longus (B) act as the antagonist muscles, while the triceps lateralis (C) and triceps longus (D) act as agonist muscles. Error bars indicate variability across participants.

During the flexion task, the agonist muscles showed increased activation as coactivation levels increased. Their activity was low during SLR but became stronger in the LLR, with the triceps longus showing the greatest increase. For triceps lateralis, the stretch activity is approximately 1

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

nu at 0.5x and 4.5 nu at 16x in SLR, and 4 nu at 0.5x and 7.5 nu at 16x in LLR. Similarly for triceps longus, the activity is about 1 nu at 0.5x and 3 nu at 16x in SLR, and 5 nu at 0.5x and 10 nu at 16x in LLR. In contrast, the antagonist muscles, the brachioradialis and biceps longus, were more inhibited as coactivation increased. Their activity decreased from the SLR to the LLR phase, meaning they contributed less to resisting the flexion movement. The brachioradialis went from about 0 nu at 0.5x to -1 nu at 16x in SLR, and 1.5 nu at 0.5x to -3 nu at 16x in LLR. Similarly for biceps longus, the stretch activity went from 0 nu at 0.5x to -7 nu at 16x in SLR, and 0 nu at 0.5x to -15 nu at 16x in LLR. The 16x coactivation level had the largest error bars, but the overall trend remained consistent where the agonist muscles became more active with higher coactivation levels, while antagonist muscles were more suppressed, especially in the later phase (LLR) of the response.

Metabolic Cost and Performance Across Coactivation Levels

Error rates and metabolic cost were measured to assess performance and energy expenditure across coactivation levels. These data are presented in a single figure combining both metrics.

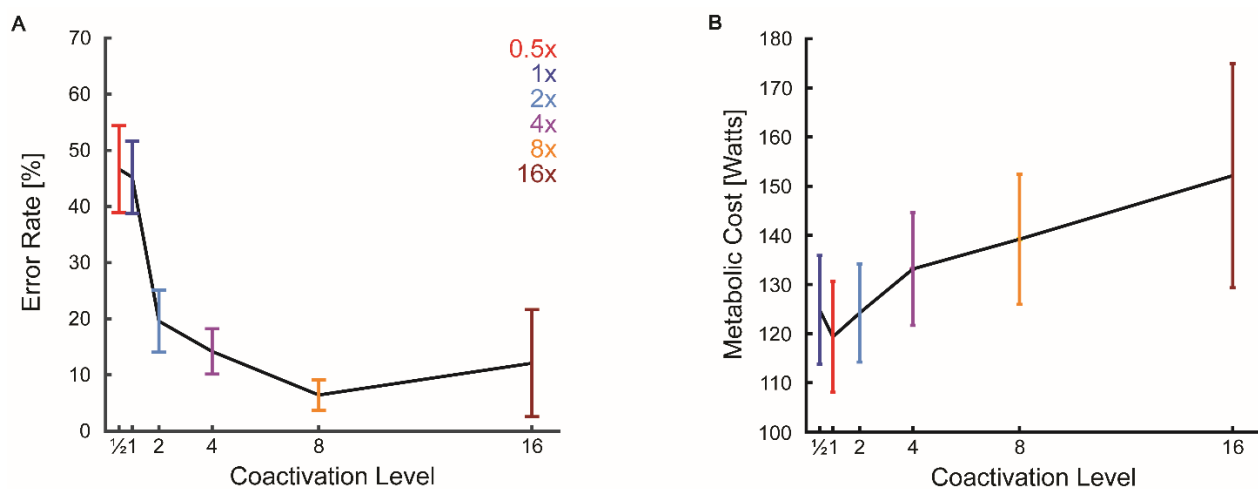


Figure 9. High coactivation improves performance (A) but increases metabolic cost (B). The color scheme was same as previous figures. The error bars indicate variability in individual responses.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

Error rates were highest (low performance) at the lowest coactivation levels, with approximately 48% error at 0.5x and around 46% at 1x. As coactivation increased, error rates dropped significantly, reaching about 15% at 4x and the lowest error rate of approximately 8% at 8x. However, at the highest coactivation level (16x), error rates increased again to around 13%, showing a non-linear trend. The error bars also showed a similar pattern, starting with large error bars at low coactivation level and then getting smaller till 8x. At the highest coactivation level, the error bars are the largest (approximately $\pm 18\%$). The overall pattern suggests that moderate to high muscle coactivation improved task performance (reduces error rate) in disturbance response before declining at extreme levels. Similarly, higher muscle coactivation levels lead to increased metabolic cost. At lowest coactivation levels (0.5x), the metabolic cost was about 125 W, while at the self-selected level (1x) the metabolic cost was lower, averaging about 120 W. However, as coactivation further increases, the metabolic cost rose steadily, reaching approximately 125 W at 2x, 133 W at 4x, 140 W at 8x and 150 W at 16x. The overall trend suggests a trade-off, that while increasing coactivation improves performance, it also results in higher energy expenditure.

Table 1. Exponential Model Fit Results across Various Conditions and Coactivation Levels (0.5, 1, 2, and 4x)

Condition	r-squared	SSE	a	b	CI
Flexion Average Elbow Angle	0.86	0.04	2.62	-0.05	(1.93, 3.30)
Average Movement Time	0.80	2011	448.20	-0.05	(292, 604.3)
Extension Average Elbow Angle	0.61	4.45	-14.16	-0.05	(-21.43, -6.90)
Error Rate	0.78	201.65	59.07	-0.19	(-5.71, 123.80)
Metabolic Cost	0.85	14.95	117.34	0.02	(105.71, 128.99)

Note. SSE = Sum of Squared Errors; CI = Confidence Interval.

a, and b are the variables in the general exponential model (equation 1).

Coactivation levels 8x and 16x were excluded.

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

The exponential model fit showed strong relationships between coactivation levels and kinematics. The flexion average elbow angle had an r-squared value of 0.86, meaning the model explained 86% of the variance. The average movement time also had a good fit ($r^2 = 0.80$), showing a strong link between coactivation and how fast participants moved. The extension average elbow angle had a lower fit ($r^2 = 0.61$), meaning coactivation explained less of the variability in movement. The error rate model had an r-squared of 0.78, consistent with the trend seen in the graph, where errors dropped with moderate coactivation but increased at the highest levels. Finally, the metabolic cost had a strong r-squared value of 0.85, meaning the model explained 85% of the variance.

The sum of squared errors (SSE) was lowest for flexion average elbow angle (0.04), meaning the model fit was the most precise for this variable. The error rate had an SSE of 201.65, while the average movement time had the highest SSE (2011), suggesting more variability in timing across participants. The extension average elbow angle had a SSE (4.45), reinforcing that coactivation had less predictive power for this condition.

In the exponential model equation, 'a' represents the initial value, and 'b' determines the rate of change. For flexion average elbow angle, $a = 2.62$ and $b = -0.05$, meaning the elbow angle started at 2.62 nu and gradually decreased. Error rate had $a = 59.07$ and a more negative $b (-0.19)$, showing a sharp initial decline in errors with increasing coactivation. The confidence intervals (CI) show how precise these estimates are. For flexion average elbow angle, the CI was (1.93, 3.30), meaning the true value of 'a' likely falls within this range. The error rate had a much wider CI (-5.71, 123.80), reflecting more variability.

Discussion

Study Overview and Broad Findings

This study questions the traditional view of coactivation as a purely mechanical strategy. It proposes that coactivation helps the nervous system with processing sensory information and controlling movement. An understanding of how coactivation balances metabolic costs and performance provides insights into motor control. The main idea was that higher muscle coactivation would improve performance but increase energy costs, while lower levels save energy but hurt performance. The results support this idea, showing that while spontaneous coactivation (1x) helped save energy, increasing muscle coactivation using biofeedback improved performance, which also used more energy. This suggests that coactivation helps adjust how we respond to sensory feedback.

Elbow Angle Dynamics

The changes in elbow angles shown in Figures 2 and 3 suggest that increasing coactivation reduces how much the arm moves when disturbed. Figure 2 tracks the angle over time, dropping from -14.5 degrees at 0.5x to -5 degrees at 16x in extension, and from 13 degrees to 5 degrees in flexion. Figure 3 confirms these peak displacement value, and shows that as muscle coactivation increases, the elbow returns closer to its original position with less overall movement. This suggests that coactivation plays an important role in controlling limb stability and reducing deviations from the circular target. This fits with Maurus et al. (2024), who argue coactivation strengthens feedback control to keep limbs steady, pushing against the old belief that muscle coactivation only stiffens muscles to resist force. Our results support the idea that coactivation helps fine-tune motor responses by limiting excessive motion. The decrease in movement with higher coactivation

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

indicates that participants were able to change their stiffness to maintain better control. This is important because it provides information to understand how coactivation balances performance and metabolic cost under disturbances.

Movement Time Optimisation

The data in Figure 4 shows how muscle coactivation affects how quickly you can return back to the target. The figure shows a decrease from 420 milliseconds at 0.5x to 280 milliseconds at 8x, then an increase to 340 milliseconds at 16x. The trend indicates that increased coactivation generally reduces movement time. This pattern builds on Maurus et al. (2023), who note the nervous system adjusts sensorimotor gains for quick responses, challenging the idea that less coactivation always enhances mobility. On the contrary, figure 4 shows that too much coactivation (16x) seems to slow things down, probably because it results in increased stiffness or metabolic constraints. This could also be explained because the error bars are the widest at 16x (± 250 milliseconds), which suggests there's a lot of variation at that level, likely due to the small number of samples that completed the 16x condition. The lowest movement time at 8x hints at an ideal spot, which could really help when training for fast recovery tasks. It brings up the interesting balance between speed and energy use when it comes to coactivation.

Muscle Activity Patterns in Extension

Figure 5 shows that as coactivation increases, the activity in the brachioradialis and biceps longus (both agonist muscles) goes up. This increase is more noticeable during the LLR window than during the SLR window (see Figure 6). Both the figures also show that antagonist muscle inhibition (triceps lateralis and triceps longus) also increased with coactivation. Greater inhibition of antagonist muscles suggests improved reciprocal inhibition, reducing opposition to the desired

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

movement (Geertsen et al., 2011). The overall trend in figures 5 and 6 was that coactivation boosts agonist activity while reducing antagonist engagement. This supports Cluff and Scott (2013) idea that coactivation can help with limb stability. They also challenge the old view that coactivation is just about resisting movement, suggesting it plays a role in enhancing feedback from the nerves. The findings imply that greater coactivation helps produce more force while also adjusting reflexes, which could lead to better stability and precision during extension tasks.

Muscle Activity Patterns in Flexion

Like the extension protocol, the flexion protocol (Figure 7 and 8) indicates that with greater coactivation, the activity of the agonist muscle increases while the antagonist muscle is more inhibited. The main difference here is which muscles are acting as agonists and antagonists. In Figures 7 and 8, you can see that as coactivation goes up, the activity in the triceps lateralis and triceps longus (the agonist muscles) also increases, and this increase is more noticeable during the LLR window than during the SLR window. On the contrary, the antagonist muscle had the most inhibition during LLR (e.g., dropping to -15 μ for biceps in Figure 8B), might reflect a physiological delay in antagonist relaxation, possibly due to cortical feedback adjustments, as suggested by Maurus et al. (2024). This builds on Molina-Rueda et al. (2023), linking coactivation to stability but suggesting that we should also think about feedback adjustment. This is important because it offers insights for flexion-specific area of how muscle coactivation increases responsiveness to proprioceptive feedback.

Performance and Metabolic Cost Trade-Off

The trends in Figure 9 shows a clear trade-off between performance (decrease in error rate) and metabolic cost as muscle coactivation increases. The figure plots error rates dropping from 48% at

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

0.5x to 8% at 8x, then rising to 13% at 16x, and metabolic cost climbing from 120 to 150 Watts. This aligns with Peterson and Martin (2010), who note coactivation improves stability but raises energy use, yet challenges the idea that less coactivation always conserves energy for mobility. The higher metabolic cost at 0.5x (125 Watts) compared to self-selected 1x (120 Watts) might stem from compensatory muscle recruitment to maintain stability, as Kistemaker et al. (2006) suggest, requiring extra effort below the natural efficiency point. This finding underscores the physiological trade-off between stability and metabolic efficiency, which is crucial for understanding motor control strategies in both everyday movements and specialized tasks. In addition, at the highest coactivation level, the error bars for error rate were the largest, approximately $\pm 18\%$ (Figure 9A), which suggests there's a lot of variation at that level, likely due to the small number of samples that completed the 16x condition. However, the overall trend suggests that as muscle coactivation increases, error rates decrease but metabolic cost increase which links back to the question of how coactivation trades off performance and metabolic cost.

Exponential Model Insights

The exponential fits in Table 1 indicate that coactivation effectively models metabolic cost, movement time, elbow angle and error trends. The table lists r^2 values from 0.61 to 0.86, SSE from 0.04 to 2011, and wide confidence intervals (e.g., -5.71 to 123.80 for error rate), with flexion showing the strongest fit (r^2 0.86). This builds on Kistemaker et al. (2006), who propose coactivation optimises control, challenging the stiffness-only perspective. The high SSE for movement time (2011 ms) reflects variability. This is important because it supports predictive models for performance optimisation, connecting to the trade-off question in dynamic tasks. The exponential model, as described by Ting and Macpherson (2005), captures these trends by

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

adjusting parameters a and b to reflect initial values and decay rates, enhancing our understanding of motor control dynamics.

Limitations of the Study

The small group of 12 participants limits generalizability across diverse populations. Some could not finish all conditions, causing wider error bars and lower r squared scores for a few measures, showing uneven data. We did not track fatigue or prior exercise, which might change EMG and metabolic results. Also, not looking at brain parts like the primary motor cortex or cerebellum, as Maurus et al (2023) mention, leaves the sensory role of coactivation unclear. Additionally, the exclusion of 8x and 16x in Table 1 due to few trials reduces understanding of extreme coactivation effects.

Future Implication and Direction

These findings may have significant implications for rehabilitation. Individuals with motor deficits, such as stroke survivors or those with multiple sclerosis, often adopt increased coactivation levels as a compensatory technique (Molina-Rueda et al., 2023). Since our findings found that coactivation improves sensory feedback, rehabilitation efforts may shift toward optimizing sensory processing rather than simply increasing muscle stiffness. To understand the range of scenarios in which coactivation can be useful, research should focus on a larger of populations, which includes older adults, athletes, and persons with impairments. Also, looking into how coactivation relates to things like muscle fatigue can help us understand motor control better.

Conclusion

We aimed to figure out how muscle coactivation shows a trade-off between performance and metabolic cost in upper limb posture control with mechanical disturbances. Our findings show that at spontaneous coactivation levels, participants tend to prioritize reducing metabolic cost at the expense of performance. However, increasing coactivation through biofeedback improves performance but also raises metabolic cost. This emphasizes the role of muscle coactivation in balancing sensory feedback. In summary, our preliminary results suggest a complex relationship between metabolic cost and performance in controlling arm movements and postures. This research provide a new framework for understanding why the nervous system coactivates skeletal muscles, which may allow for better treatments in a clinical setting for excessive muscle tone (muscle spasticity) and abnormally sensitive reactions to sensory feedback (hyperreflexia).

Acknowledgement

This work was supported by grants from the Natural Sciences and Engineering Research Council of Canada (NSERC). We thank Dr. Tyler Cluff and members of the Integrative Sensorimotor Neuroscience Lab of the University of Calgary for insightful comments and suggestions.

Philipp Maurus was primarily affiliated with the University of Calgary at the time he assisted with this project but is currently affiliated with the National Institute of Neurological Disorder and Stroke (NINDS).

Author contributions

Conceptualization, P.M., and T.C.; Methodology, B.D., and T.C.; Software, P.M., and T.C.; Formal Analysis, B.D., and J.R.; Investigation, B.D., and P.M.; Writing – Original Draft, B.D.; Writing – Review & Editing, B.D., and T.C.; Funding Acquisition, T.C.; Supervision, T.C.

521 **Declaration of interests**

522 The authors declare no competing interests.

523 **Inclusion and diversity**

524 We support inclusive, diverse, and equitable conduct of research.

525

526

527

528

529

530

531

532

533

534

535

536

537

538

References

- Balshaw, T., Massey, G., Maden-Wilkinson, T., Lanza, M., & Folland, J. (2018). Neural adaptations after 4 years vs 12 weeks of resistance training vs untrained. *Scandinavian Journal of Medicine and Science in Sports*, 29(3), 348-359.
<https://doi.org/10.1111/sms.13331>
- Brockway J. M. (1987). Derivation of formulae used to calculate energy expenditure in man. *Human nutrition. Clinical nutrition*, 41(6), 463–471.
- Cluff, T., & Scott, S. H. (2013). Rapid feedback responses correlate with reach adaptation and properties of novel upper limb loads. *The Journal of neuroscience : the official journal of the Society for Neuroscience*, 33(40), 15903–15914.
<https://doi.org/10.1523/JNEUROSCI.0263-13.2013>
- Gebel, A., Lüder, B., & Granacher, U. (2019). Effects of increasing balance task difficulty on postural sway and muscle activity in healthy adolescents. *Frontiers in Physiology*, 10.
<https://doi.org/10.3389/fphys.2019.01135>
- Geertsen, S. S., Stecina, K., Meehan, C. F., Nielsen, J. B., & Hultborn, H. (2011). Reciprocal Ia inhibition contributes to motoneuronal hyperpolarisation during the inactive phase of locomotion and scratching in the cat. *The Journal of physiology*, 589(Pt 1), 119–134.
<https://doi.org/10.1113/jphysiol.2010.199125>
- Kesar, T. M., Tan, A., Eicholtz, S., Baker, K., Xu, J., Anderson, J. T., Wolf, S. L., & Borich, M. R. (2019). Agonist-Antagonist Coactivation Enhances Corticomotor Excitability of Ankle Muscles. *Neural plasticity*, 2019, 5190671. <https://doi.org/10.1155/2019/5190671>

POSTURAL COACTIVATION SHOWS COST-PERFORMANCE TRADE-OFF

- 560 Kistemaker, D. A., Van Soest, A. J., & Bobbert, M. F. (2006). Is equilibrium point control
561 feasible for fast goal-directed single-joint movements?. *Journal of*
562 *neurophysiology*, 95(5), 2898–2912. <https://doi.org/10.1152/jn.00983.2005>
- 563 Maurus, P., Armstrong, D. P., Scott, S. H., & Cluff, T. (2024). Muscle coactivation primes the
564 nervous system for fast and task-dependent feedback control. *ArXiv (Cornell University)*.
565 <https://doi.org/10.48550/arxiv.2410.16101>
- 566 Maurus, P., Jackson, K., Cashaback, J. G. A., & Cluff, T. (2023). The nervous system tunes
567 sensorimotor gains when reaching in variable mechanical environments. *iScience*, 26(6),
568 106756. <https://doi.org/10.1016/j.isci.2023.106756>
- 569 Molina-Rueda, F., Fernández-Vázquez, D., Navarro-López, V., López-González, R., &
570 Carratalá-Tejada, M. (2023). Muscle Coactivation Index during Walking in People with
571 Multiple Sclerosis with Mild Disability, a Cross-Sectional Study. *Diagnostics (Basel,*
572 *Switzerland)*, 13(13), 2169. <https://doi.org/10.3390/diagnostics13132169>
- 573 Peterson, D. and Martin, P. (2010). Effects of age and walking speed on coactivation and cost of
574 walking in healthy adults. *Gait & Posture*, 31(3), 355-359.
575 <https://doi.org/10.1016/j.gaitpost.2009.12.005>
- 576 Ting, L. H., & Macpherson, J. M. (2005). A limited set of muscle synergies for force control
577 during a postural task. *Journal of neurophysiology*, 93(1), 609–613.
578 <https://doi.org/10.1152/jn.00681.2004>
- 579 Vieira, H., Ferraz, T., Sousa, T., Martins, I., Marques, A., & Reis, S. (2019). Study of the risk of
580 ankle injury during impact on the ground and definition of support orthoses..
581 <https://doi.org/10.1109/enbeng.2019.8692518>